

Learning Location, Not Looking: A Placement-Memorization Audit and Ten-Episode Repair at the 30M Scale

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Abstract

Fixed-placement benchmarks let imitation policies learn *location* instead of *looking*. We document this on LIBERO-Object, whose target start poses are exactly pinned (measured) in 6 of 10 tasks and within $\sim \pm 1$ cm in the rest. First, an audit finds placement memorization at three layers: a calibrated expert whose offset constant encodes pose (sign-flipping under a software rebuild), an iteration-coupled selection loop (0.700 dev vs. 0.300 held-out), and a learned grasp head whose regression rationally prefers proprioception to pixels. Second, a repair—ten teacher episodes under ± 4 cm placement teleports plus jitter on a nuisance input—restores grounding (attribution: vision 1.1–1.8 cm, proprioception 0.1 cm), verified by a substitution probe paired with a behavioural randomization protocol: the released head scores 0.400 dev, 0.700 held-out, 0.400 randomized, with both non-dev protocols confirmed at $n=50$ (35/50 [0.56, 0.81]; 26/50 [0.39, 0.65]). The protocol’s twin control cells scope that verification from both sides: on the deployment stack the memorized head scores 1/10 under the repaired head’s own shift draws (where it scores 4/10), while on a rebuilt stack whose detector shift disables the repaired head’s visual goals (0/10), the shell’s search absorbs the same shifts for the proprioception-shortcut head (6/10). The behavioural certificate is therefore joint in (head, stack). Third, the policy class that made this reachable is goal-structured decoding driving an engineered pick-and-place shell; free per-tick regression scored 0/56. All results are simulation, one task and one object in every claimed cell, $n=10$ per cell and final at that n . A dated addendum carries the multi-object campaign: four objects attempted, a second crossed and twice confirmed on never-scored seeds, two measured, mechanism-located at role binding, and *not* crossed—with two pre-registered predictions recorded and falsified, and four instruments built to quantify the boundary all abandoned at their own calibration gates.

1 Introduction

A policy that scores 0.700 on a manipulation benchmark has, on the face of it, learned to see, decide, and act. This paper is about how much of such a number can instead be *location*: constants—in a benchmark’s init files, in an expert’s calibration, in a learned regressor’s shortcut—that encode where the object always is, so that vision only ever gates an approach a memorized value finishes. On LIBERO-Object [9] we measure the preconditions exactly: the target object’s start pose is bit-identical across all 50 shipped init states in 6 of 10 tasks, and varies by only 0.25–0.58 cm (std) in the other four. A benchmark with $\lesssim 1$ cm of target variance admits position-memorizers, and we found them in every layer of our own system.

The vehicle is MicroVLA, a deliberately small language-conditioned vision-language-action stack: ~ 30 M parameters deployed, with a frozen open-vocabulary detector serving as the only vision *and* text encoder. The size is the paper’s setting, not its claim. Its methodological value is that a stack this small has no capacity to paper over systems defects, so every seam had to be instrumented—and the instruments are what this paper contributes.

Contributions.

1. **A placement-memorization audit and its verification method.** A forensically worked instance of shortcut learning [5, 6] on a fixed-placement benchmark, shown at three layers, with the corrected measured account of what LIBERO-Object pins (per-task table and shipped measurement script); and

the probe-plus-behavioural-randomization protocol, motivated by the on-manifold limit of substitution probes and by a counterexample to probe-only certification. The protocol’s twin control cells sharpen its scope: behavioural randomization separates memorized from repaired heads under identical draws on the deployment stack (1/10 vs. 4/10), and fails to separate them on a stack whose rebuild disables visual goals (6/10 vs. 0/10).

2. **A controlled policy-class result.** With perception, trunk, and corpus regime fixed, free per-tick regression scores 0/56 (upper 95% bound 6.4%) while goal-space supervision driving an engineered pick-and-place shell scores 4/10 dev, 7/10 held-out, 4/10 randomized—claimed strictly as a bundle, with the shell’s share measured on both stacks.
3. **Training–serving-skew instrumentation.** 29 worked defect instances at producer/consumer seams, split into disagreements catchable by parity testing and agreements-on-a-wrong-convention catchable only by provenance [3, 13].

2 Platform

Table 1: Deployed parameter ledger. “Deployed” counts every parameter loaded at inference, frozen or trained.

component	params	status
YOLO-World-S detector (vision <i>and</i> the only text encoder)	13.0 M	frozen
Tiny Recursive Model world model	9.97 M	frozen at deployment
fusion + drift + relational + planner trunk heads	7.0 M	trained (≤ 9 M cap)
goal heads (grasp 0.17 M + place 0.07 M) + learned gates	0.24 M	trained
deployed total	≈ 30 M	

No separate language model exists anywhere in the stack: the command is parsed into source/target phrases, and CLIP embeddings for (command, source, target) are harvested once per task from the detector’s own text tower. **The world model is causally inert in every nonzero number of this paper**—a persistence-TRM ablation reproduces the held-out cell exactly (0.700).

3 Results

Table 2: Protocol $\times$ head. All cells $k/10$ with Wilson 95% intervals; task 0, wrist camera. Audit-stack cells are never pooled with deployment-stack cells (§ stack finding).

protocol	memorized head	flagship (released)	sibling A	sibling B
dev (states 0–9)	7/10	4/10	7/10	7/10
held-out (states 10–19)	3/10	7/10	7/10	6/10
randomized ± 4 cm (same draws)	1/10	4/10	5/10	3/10
randomized ± 4 cm (audit stack)	6/10	0/10	—	—

Confirmed at power: held-out **35/50** = 0.700 [0.56, 0.81]; randomized **26/50** = 0.520 [0.39, 0.65]. Across a detector-stack rebuild the repaired head reproduces in all three protocols while the memorized head’s dev cell collapses 7/10 $\rightarrow$ 2/10 to its held-out floor: memorization is stack-coupled, grounding is what survived.

4 The multi-object addendum, and a falsified prediction

A second object (butter) crosses: 10/10 under an early-latch configuration, 6/10 with soup at 4/10 under a single anchor-band configuration with no per-object constants, and two independent pre-registered fresh-seed confirmations (5/10 + 3/10; 5/10 + 4/10 on a three-object head). Two further objects (cream cheese,

chocolate pudding) do not cross, and the mechanism is located rather than asserted: goals are accurate where measurable (cream 1.58 cm on-manifold, vs. 1.34 and 1.06 for the objects that work) while *role binding* fails identically on both — each teacher never leaves its visual-servo phase, at source-uv std ~ 0.30 , settling tens of centimetres from the target. Four objects were attempted; two crossed.

Table 3: Role-binding sub-study: four binders, three objects, $n=10$ each, one head and one configuration. Offline identity accuracy is leave-episode-out 3-way classification on corpus crops.

binder	offline identity acc.	cream	butter	soup
none	—	0/10	4/10	5/10
crop-CLIP rerank	not measured	0/10	7/10	5/10
mean prototype (centered)	0.613	0/10	2/10	8/10
1-NN bank (centered)	0.902	0/10	4/10	6/10

Offline binder accuracy dissociates from deployed success. The weaker binder owns the best soup cell and the worst butter cell; the stronger returns butter to baseline; the binder with no offline number wins butter. A matched-episode A/B probe explains it: with the 0.902 binder active the uv stream fed to the grasp head *destabilises* (std $0.258 \rightarrow 0.335$ laterally, $0.047 \rightarrow 0.202$ vertically)—a binder accurate on corpus crops thrashes on live crops. This is the paper’s own on-manifold limit recurring on two further instruments: a probe that evaluates on teacher trajectories cannot certify off-manifold behaviour, and a binder *built* from teacher trajectories cannot bind off-manifold crops.

Two predictions were pre-registered here and both are recorded as falsified: that 0.82 per-tick identity accuracy would yield 0.914 median-of-3 latch correctness and hence cream > 0 under bank binding (the cell returned 0/10), and that chocolate pudding would cross because it rarely won a misbind (its teacher never reached a grasp in five calibration iterations). Quantifying the deployed side of this boundary was attempted four times and abandoned four times, each at a pre-registered calibration gate: a projected-origin ground truth (rejected — soup, which succeeds 35/50, scored 0.111), its sign-corrected successor (rejected — the in-frame filter deletes the close-up target by construction), and three text-tower discrimination probes (rejected — each failed to detect the working object’s own phrase). Deployed binding accuracy and text-tower discriminability are therefore reported as **unmeasured**, not estimated. The error is attributable—the prediction assumed offline per-tick accuracy equals deployed per-tick accuracy, which the probe refutes and which we do not measure. The prediction, its falsification, and this attribution are retained rather than removed.

5 Related work

Scale and size are comparable only under a stated convention; under “deployed = every parameter loaded at inference”, RT-2 [4] is $\sim 55\text{B}$, OpenVLA [8] $\sim 7\text{B}$, SmolVLA [15] $\sim 450\text{M}$, Octo-small [12] $\sim 137\text{M}$ with its frozen text encoder, and MicroVLA $\approx 30\text{M}$. We claim no comparative superlative, and no number here is comparable to community-protocol LIBERO scores (600 steps, $n=10$, wrist-only, scripted-expert corpora). Predicting *where* rather than *how* is the Transporter [17] and CLIPort [14] lineage, continued by keypoint interfaces over engineered primitives [7, 10] and, with both levels learned, $\pi_{0.5}$ [2]. Our audit is a worked manipulation instance of causal confusion and copycat shortcuts [1, 5, 16]; what we add is the audited counterexample to probe-only certification, the resulting probe-plus-randomization pairing, and the control cells that scope the pairing itself. Bootstrapping from scripted experts is established [11]; our delta is that the scripted teacher is itself audited, caught encoding placement and detector version, and repaired.

6 Claims, non-claims, limitations

Claimed: the pinning measurement in its scoped form; the released head’s cells above; free per-tick regression at 0/56; the de-memorization mechanism at the head level (attribution shift, varied-label validation, and a paired jitter ablation, $0/10 \rightarrow 7/10$ dev, exact McNemar/sign $p \approx 0.016$), scoped by the audit-stack control; and 29 worked skew defects. **Not claimed:** any task-success contribution from the world model (causally inert in every nonzero number); end-to-end learned motor control (the shell is engineered); head-level

grounding from the randomized column alone, stack-free; object-level generalization beyond the addendum’s two crossed objects (four attempted, two blocked at role binding); any deployed binding-accuracy figure; any comparative size superlative; physical deployment. **Limitations:** $n=10$ per cell with $n=50$ confirmations on two protocols; ten held-out states reused across five disclosed selection looks; machine constants inherited from a dev-tuned era; substitution probes are on-manifold instruments; ± 4 cm randomization sits inside the shell’s ± 6 cm probe envelope and separates heads only where visual goals are live; the teacher’s calibration consumed privileged diagnostic telemetry offline; single benchmark, single simulator.

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